

Continuity and the Intermediate Value Theorem



The three-part continuity test

- 1 State the three conditions that must all hold for f to be continuous at $x = a$.
 - 2 Determine whether $f(x) = \frac{x^2 - 4}{x - 2}$ (for $x \neq 2$, with $f(2) = 4$ defined separately) is continuous at $x = 2$.
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Finding values that ensure continuity

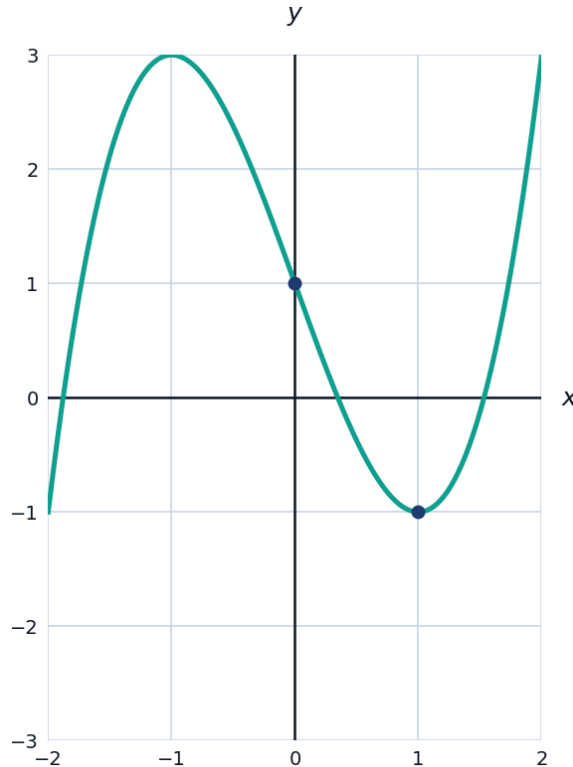
- 3 Find k so that $f(x) = kx^2$ for $x \leq 2$, and $f(x) = 5x - 2$ for $x > 2$, is continuous at $x = 2$.
 - 4 Find a and b so that $f(x) = 2$ for $x \leq -1$, $f(x) = ax + b$ for $-1 < x < 2$, and $f(x) = 6$ for $x \geq 2$, is continuous everywhere.
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Types of discontinuity

- 5 Classify the discontinuity of each function at the indicated point:
 - a $f(x) = \frac{x+3}{x^2-9}$ at $x = -3$
 - b $f(x) = \lfloor x \rfloor$ at $x = 2$
 - c $f(x) = \frac{1}{(x-1)^2}$ at $x = 1$
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The Intermediate Value Theorem

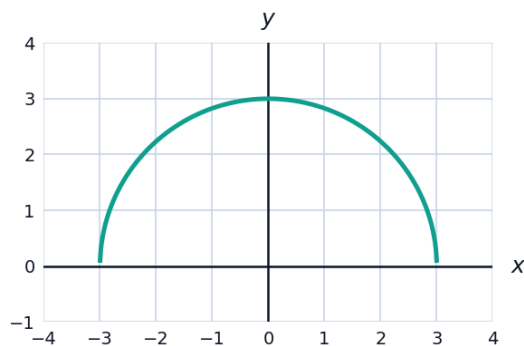
- 6 Show that $f(x) = x^3 - 3x + 1$ has a root in $(0, 1)$, using the IVT.



- 7 A continuous function g satisfies $g(1) = -3$ and $g(5) = 7$. Explain why g must take the value 0 somewhere in $[1, 5]$, and state the theorem used.
- 8 A cup of coffee's temperature (in $^{\circ}\text{C}$) is a continuous function of time, starting at 90°C and cooling to 25°C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly 50°C at some point.

Continuity on an interval

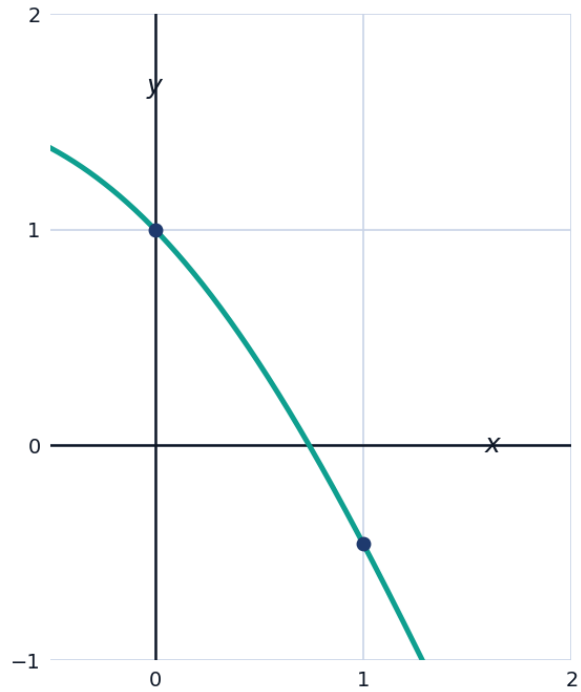
- 9 Determine the largest interval(s) on which $f(x) = \sqrt{9 - x^2}$ is continuous.



- 10 Determine all values of x where $f(x) = \frac{x+1}{x^2-x-6}$ is discontinuous, and classify each discontinuity.

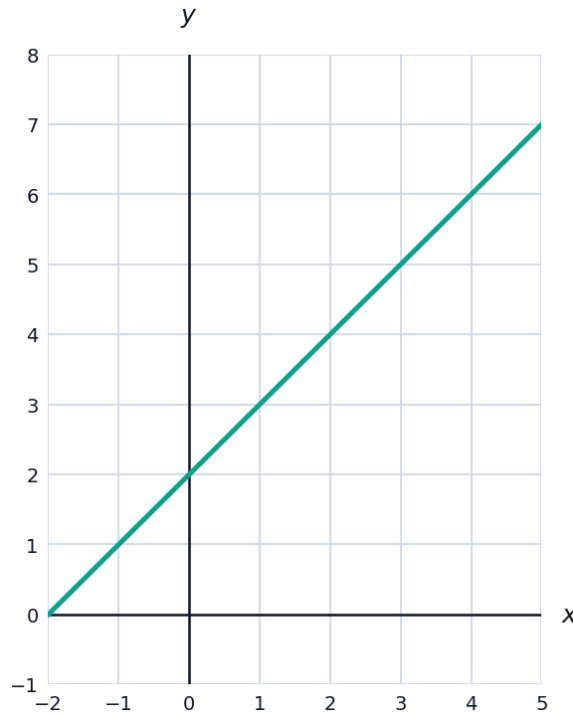
Using the IVT to approximate roots

- 11** Show that $f(x) = \cos x - x$ has a root in $(0, 1)$, then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.



Visualizing a removable discontinuity

- 12 The graph shows $f(x) = \frac{x^2 - 4}{x - 2}$ for $x \neq 2$, simplifying to $f(x) = x + 2$. Use the graph to describe what would need to be true of $f(2)$ for the function to be continuous there.



Continuity of piecewise and combined functions

- 13 Determine whether $f(x) = |x - 3|$ is continuous at $x = 3$, checking all three conditions.
- 14 If f and g are both continuous at $x = a$, explain why $f(x) + g(x)$ and $f(x) \cdot g(x)$ must also be continuous at $x = a$, citing the relevant limit laws.

An applied IVT problem

- 15 A hiker's elevation (in meters) is a continuous function of distance walked, starting at 200 m and ending at 850 m after 5 km. Explain, using the IVT, why the hiker must have been at exactly 500 m at some point, and state why this reasoning would fail if elevation were not continuous.

Continuity with absolute value and radical functions

- 16 Determine the domain (and hence largest interval of continuity) of $f(x) = \sqrt{x^2 - 4}$.